Answer Key

Official

JOB ID: demo-full-structured TOTAL QUESTIONS: 7 TOTAL MARKS: 33

CONFIDENTIAL

Name: _____ Roll No.: _____ Date: _____

Question 1 **[5 Marks]**

Identify the methods used to solve a system of linear equations, including substitution, elimination, and matrix methods.

correct_option	**1 Mark**

Question 2 **[10 Marks]**

Recall that the derivative of sin(x) with respect to x is cos(x).

correct_option	**1 Mark**

Question 3 **[10 Marks]**

Identify that the derivative at a point provides the slope of the tangent line to the graph at that point, reflecting the instantaneous rate of change.

correct_option	**1 Mark**
explanation	**The correct answer is (b) because the derivative at a specific point measures the slope of the tangent line to the function's graph at that point, representing the instantaneous rate of change. Marks**

Question 4 **[2 Marks]**

1. Apply the Pythagorean theorem: c^2 = a^2 + b^2. 2. Substitute a = 6 cm and b = 8 cm: c^2 = 6^2 + 8^2. 3. Calculate squares: c^2 = 36 + 64. 4. Sum the squares: c^2 = 100. 5. Take the square root of 100: c = √100. 6. Final answer: c = 10 cm.

step_1	**0.5 Marks**
step_2	**0.5 Marks**

Question 5 **[2 Marks]**

1. Identify the known values: one leg = 9 cm, hypotenuse = 15 cm. 2. Apply Pythagorean theorem: a^2 + b^2 = c^2. 3. Substitute known values: 9^2 + b^2 = 15^2. 4. Calculate squares: 81 + b^2 = 225. 5. Solve for b^2: b^2 = 225 - 81 = 144. 6. Find b: b = √144 = 12 cm.

step_1	**0.5 Marks**
step_2	**0.5 Marks**
step_3	**0.5 Marks**
step_4	**0.5 Marks**
step_5	**0.5 Marks**
final_answer	**0.5 Marks**

Question 6 **[2 Marks]**

Apply the Pythagorean theorem: a^2 + b^2 = c^2. Substitute known values: 5^2 + b^2 = 13^2. Calculate 5^2 = 25 and 13^2 = 169. Subtract 25 from 169 to find b^2: 169 - 25 = 144. Take the square root of 144 to find b: √144 = 12. Final answer: the other leg measures 12 cm.

correct_application	**1 Mark**
correct_substitution	**0.5 Marks**

correct_calculation	**0.25 Marks**
final_answer	**0.25 Marks**

Question 7 **[2 Marks]**

State the form ax + b = 0; explain that a and b are constants, with 'a' being the coefficient of the variable and 'b' the constant term.

correct_form	**1 Mark**
correct_explanation	**1 Mark**

*** END OF ANSWER KEY ***

Examiner's Signature